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← Mechanics →

$$r = R + h$$

$$v = \sqrt{g \cdot r}$$

$$T = \frac{2\pi r}{v}$$

$$P.E = mgh$$

$$\Delta P.E = P.E_1 - P.E_2$$

$$H_N = (0.9)^N H$$

$$g_m = \frac{GM_m}{R_m^2}$$

$$P.E = \frac{GMm}{r}$$

$$K.E = \frac{1}{2} m v^2$$

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$$V_{esc} = \sqrt{\frac{2Gm}{R_m}}$$

$$\vec{v} = \frac{d}{dt} \vec{r}$$

$$\vec{p} = m\vec{v}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$F_g = \frac{GMm}{r^2}$$

$$F_N = \frac{mv^2}{r}$$

$$R = \frac{V_0^2 \sin 2\theta}{g}$$

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$$\theta' = 90 - \theta$$

$$T = \frac{2V_0 \sin \theta}{g}$$

$$t = \frac{V_0 \sin \theta}{g}$$

$$X = V_{0x} t \Rightarrow t = X/V_{0x}$$

$$V_x = V_{0x} = V_0 \cos \theta$$

$$V_y = V_0 \sin \theta$$

$$y = V_{0y} t - \frac{1}{2} g t^2$$

$$V_y = V_{0y}, \quad V_x = V_{0x}$$

$$V_y = V_{0y} - g t$$

$$X_p = V_{0x} t$$

$$V_0 = \sqrt{(V_{0x})^2 + (V_{0y})^2}$$

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$$\theta = \tan^{-1} \frac{V_{oy}}{V_{ox}}$$

$$H = \frac{V_o^2 \sin^2 \theta}{2g}$$

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$$\frac{N_2}{N_1} = \frac{N_0 e^{-E_2/KT}}{N_0 e^{-E_1/KT}}$$

$$v = f \lambda$$

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

$$P = NE$$

$$E = \frac{1}{2}mv^2$$

$$\lambda = \frac{h}{mv}$$

$$\lambda_{\max} \times T = 0.2898 \times 10^{-2} \text{ mK}$$

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$$\phi = hf_0$$

$$E = hf$$

$$K.E = hf - \phi$$

$$\lambda = \frac{hc}{\phi}$$

$$\Delta\lambda = \lambda' - \lambda = \lambda_c (1 - \cos\theta)$$

$$\lambda_c = \frac{h}{m_e c} = 2.43 \text{ pm}$$

$$K.E = hc \left(\frac{1}{\lambda_i} - \frac{1}{\lambda_f} \right)$$

$$\frac{\Delta E}{E} = \frac{E_f - E_i}{E_i}$$

$$\Rightarrow E = \frac{hc}{\lambda} = \phi_0$$

$$E = \phi_0 + eV$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2$$

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$$K.E = hf - hf_0$$

$$A = 4\pi R^2$$

$$\frac{P}{A} = \sigma e T^4$$

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$P = \frac{E}{t}$$

$$\text{No. of fission} = \frac{E}{E_{\text{fission}}}$$

$$\text{Total mass} = \text{mass} \times \text{no of atoms}$$

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$$E = \frac{q}{2\pi\epsilon_0 r l}$$

$$F = Eq$$

$$2T\sin\theta - 2F_e = 0 \quad (T\sin\theta = F_e)$$

$$W = mg$$

$$\Sigma F_y = 0 \Rightarrow 2T\cos\theta = W$$

$$F_e = T\sin\theta$$

$$F_e = \frac{kq^2}{r^2}$$

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$$Ae^{-bT/2m}$$

$$E = \frac{1}{2} K x_0^2$$

$$1) K.E = \frac{1}{2} m v^2$$

$$2) \pm \frac{\Delta K.E}{K.E} = \frac{\Delta m}{m} + \frac{2 \Delta v}{v}$$

$$3) y = b e^x \quad (\text{exponentially increasing})$$

$$4) y = K (1 - e^x) \quad (-ve \text{ curve})$$

$$5) N = N_0 Z^{-\lambda t}$$

$$6) P.E = \frac{1}{2} K x^2$$

$$7) \text{Area of ellipse} = \pi xy$$

$$8) y = a e^{bx} \quad (\text{exp. func'})$$

$$9) y = a x^b \quad (\text{poly. func})$$

$$10) \int A dl = \int \text{curl } A dA$$

$$11) \nabla \times F = 0$$

$$12) F = \nabla \phi$$

$$13) -\vec{\nabla} \cdot \vec{V} = \vec{F}$$